Smart IOT-based Healthcare system for early disease detection using plasmonic PCF-SPR sensors

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ABSTRACT

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Chapter 1: Introduction

1.1 Preamble

In recent years, modern medical systems require accurate and fast technologies for disease detection and patient monitoring. Biosensors play an important role due to their ability to detect biological changes efficiently. Surface Plasmon Resonance (SPR) is widely used because of its high sensitivity and quick response. When combined with Internet of Things (IoT) technology, it enables real-time monitoring and remote data transmission .This project aims to develop a simple and effective medical monitoring system based on these technologies.

1.2 Background of the Problem

In recent years, there has been an increasing need for highly sensitive and reliable sensors in the medical field, especially for early disease diagnosis and monitoring of patients' vital signs. Traditional sensing systems face several limitations, including:

Low sensitivity to detect small changes in biological samples.

Limited sensing range, which may prevent detection of certain biomarkers.

Large system size, making it difficult to use in clinical or portable settings.

Susceptibility to environmental noise and external disturbances.

These limitations make it challenging to achieve accurate and real-time medical detection, which is essential for modern healthcare and early disease diagnosis.

- **In the following sections, the fundamental concepts of optical and electronic sensors, as well as biosensors, will be discussed to provide the necessary background for understanding advanced SPR-based sensing systems.**

1.3 Optical Sensing Technologies

Optical sensing technologies detect physical chemical and biological changes with high sensitivity. They are based on light matter interaction where changes affect optical properties such as intensity phase or wavelength .Compared to electrical sensors they provide immunity to electromagnetic interference fast response and harsh environment operation and are used in environmental biomedical and industrial applications .A key application is refractive index sensing which enables detection of changes in concentration temperature or biochemical reactions for label free sensing [1].

1.3.1 Limitations of Conventional Sensing Techniques

Conventional sensing techniques have limitations such as low sensitivity which reduces their ability to detect small changes. They also require bulky systems that increase cost and are affected by electromagnetic interference which reduces accuracy. In addition they have limited real time monitoring and often need calibration or sample preparation. These issues led to the development of optical sensing technologies [2].

1.4 Optical Sensors

Optical sensors convert light changes into measurable signals to detect physical chemical or biological parameters. They offer high sensitivity fast response and immunity to electromagnetic interference and can measure temperature pressure strain refractive index and biochemical substances. They work by changes in light properties such as intensity wavelength phase or polarization in different operating modes [3].

1.4.1 Electronic Sensors

Electronic sensors convert physical chemical or biological changes into electrical signals. They are widely used due to low cost and simple design. However they suffer from electromagnetic interference lower sensitivity and need direct contact which can affect accuracy. Despite this they remain important due to their simplicity and reliability [4].

1.4.2 Optical Sensing

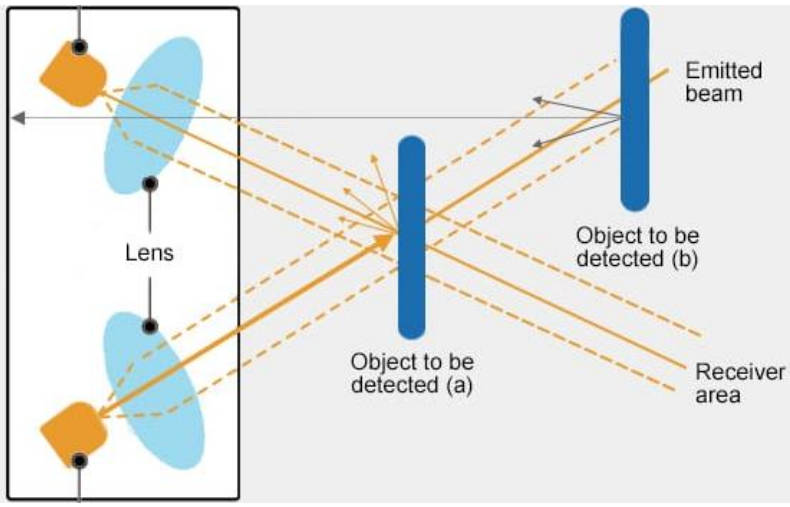

Figure 1.1: Basic principle of an optical sensor showing light interaction with the sensing element[3]

Optical sensing uses light to detect changes in physical, chemical, or biological parameters by measuring variations in light properties such as intensity, wavelength, phase, or polarization [5].It offers high sensitivity, immunity to electromagnetic interference, compact size, and supports real-time and remote monitoring. It also enables label-free detection, and advanced techniques like surface plasmon resonance, interferometry, and photonic crystal fiber sensors improve performance

1.5 Biosensors

1.5.1 Definition and Components of Biosensor

A biosensor is an analytical device that combines a biological recognition element with a transducer to convert a biological reaction into a measurable signal. It is widely used in medical diagnostics, environmental monitoring, food safety, and biotechnology due to its high sensitivity and selectivity .The biological element may include enzymes, antibodies, nucleic acids, or cells, while the transducer converts the interaction into signals such as electrical or optical outputs. Biosensor performance depends on sensitivity, selectivity, response time, and stability, enabling fast and often label-free detection [8], [6].

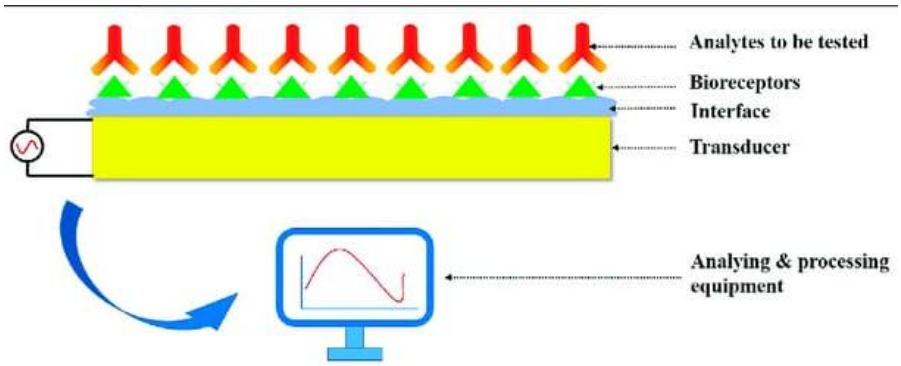

Figure 1.2: General structure of a biosensor and its main components[8]

1.5.2 Types of Biosensors

Biosensors are classified based on the biological element or transduction method and include enzymatic, immunosensors, DNA, cell-based, and optical biosensors.

Enzymatic biosensors use enzymes to generate signals related to analyte concentration. Immunosensors rely on antigen–antibody reactions. DNA biosensors detect genetic sequences. Cell-based biosensors use living cells, while optical biosensors detect changes in light properties such as in SPR systems .Each type is selected based on sensitivity, selectivity, and application needs [8].

1.5.3 Applications of Biosensors

Biosensors are used in medical diagnosis, environmental monitoring, food safety, and biotechnology.They detect glucose, pathogens, pollutants, and contaminants with high sensitivity and fast response.Recent developments like microfluidic and lab-on-a-chip systems have improved portability and automation [9].

1.6 Surface Plasmon Resonance (SPR)

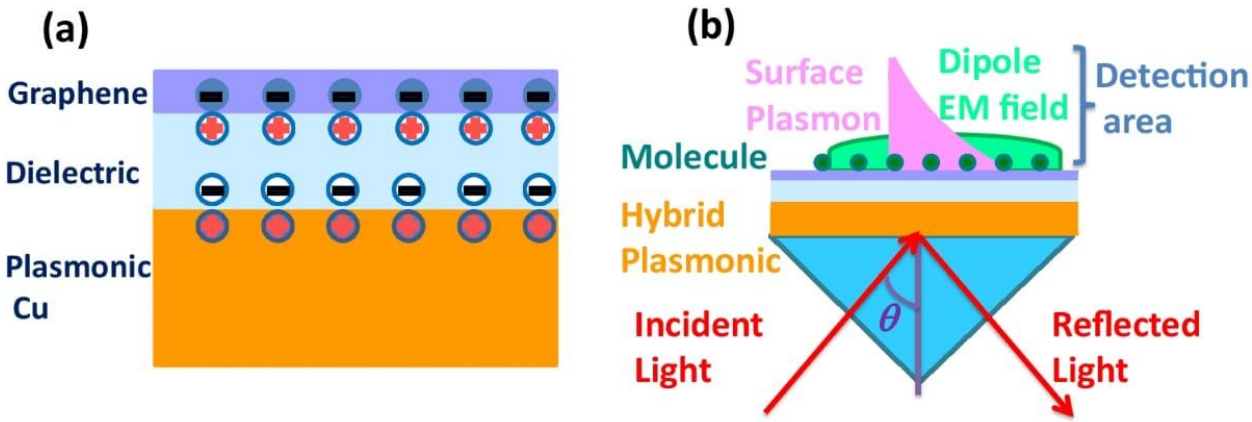

Figure1.3: Surface plasmon resonance at a metal-dielectric interface[2]

1.6.1 Definition

Surface Plasmon Resonance (SPR) is an optical phenomenon that occurs when incident light interacts with free electrons at a metal–dielectric interface, generating surface plasmons. SPR is extremely sensitive to variations in the refractive index near the metal surface, which makes it a powerful technique for real-time detection of chemical and biological substances. This high sensitivity enables label-free and rapid sensing, and has led to widespread use of SPR sensors in medical diagnostics, environmental monitoring, and biochemical analysis [1], [2].

1.6.2 Types of SPR

SPR sensors can be classified according to their excitation configuration and structural design. Prism-based SPR systems use a coupling prism to excite surface plasmons and are widely employed inlaboratory-scale, high-precision measurements. Grating-based SPR sensors utilize diffraction gratings to couple light into the plasmonic layer and are suitable for compact and portable sensing platforms. Fiber-optic SPR sensors integrate SPR with optical fibers, offering advantages such as remote sensing capability, compactness, and compatibility with harsh environments [3].

1.6.3 Features and Characteristics of SPR

Sensors possess several characteristics that make them highly effective for sensing applications, including high sensitivity to small refractive index changes, label-free detection capability, rapid response, and non-invasive optical measurement. In addition, SPR is highly versatile and can be applied in medical diagnostics, environmental monitoring, food safety, and biochemical research [1].

1.6.4 Applications of SPR

SPR technology has been extensively applied in medical diagnostics for detecting biomarkers, proteins, DNA, and pathogens. In environmental monitoring, SPR sensors are used for detecting pollutants and hazardous substances in water, air, and soil. SPR is also widely employed in food safety for contaminant detection, as well as in pharmaceutical research for studying molecular binding kinetics and drug–target interactions [2].

1.6.5 Relation with Refractive Index

The operating principle of SPR sensors is directly dependent on changes in the refractive index near the metal surface. Binding of an analyte to the sensing layer modifies the local refractive index, resulting in a measurable shift in the resonance angle or wavelength. This shift is correlated with analyte concentration, enabling highly sensitive detection of chemical and biological species [3].

1.7 Comparison between Classical and Modern SPR

Classical SPR systems are typically prism-based and provide high sensitivity but suffer from bulky size and limited portability. In contrast, modern SPR configurations incorporate optical fibers, gratings, and nanostructured materials to achieve compact, flexible, and portable sensor designs. Modern SPR sensors support remote sensing and real-time monitoring while maintaining high sensitivity and specificity, reflecting a clear trend toward miniaturization and system integration [9].

1.8Fiber-Optic SPR Sensors

1.8.1 Integration of Optical Fiber with SPR

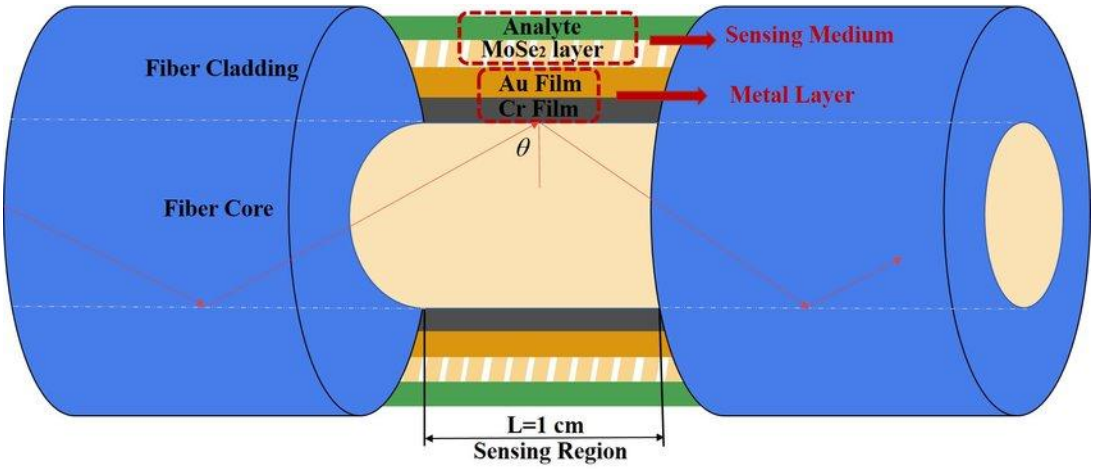

Figure 1.4: Schematic of a fiber-optic SPR sensor with metal coating[4]

Fiber-optic SPR sensors combine surface plasmon resonance with optical fibers to enable compact and flexible sensing platforms. Light propagating through the fiber interacts with a metal-coated region, exciting surface plasmons. The resonance condition is highly sensitive to refractive index changes in the surrounding medium, allowing accurate detection of chemical and biological analytes[4].

1.8.2 Metal Coating Methods

The metal layer is essential for SPR excitation in fiber-optic sensors. Common deposition techniques include physical vapor deposition (PVD), chemical vapor deposition (CVD), and electrochemical deposition. The choice of coating method strongly influences sensor sensitivity, stability, and reproducibility [4].

1.8.3 Fiber-Based SPR Sensors

Fiber-based SPR sensors offer several advantages, including miniaturization suitable for in-vivo sensing, remote detection capability, and high flexibility for integration into complex or harsh environments. These sensors are widely used in biomedical diagnostics, environmental monitoring, food safety, and chemical sensing applications [5].

1.9 Photonic Crystal Fiber (PCF)

1.9.1 Definition

Photonic Crystal Fibers (PCFs) are optical fibers characterized by a periodic arrangement of air holes along their length, enabling precise control of light confinement, propagation, and dispersion. Due to their unique guiding properties, PCFs are extensively used in sensing, nonlinear optics, and telecommunication applications [7].

1.9.2 Types of PCFs

PCFs can be classified into index-guiding PCFs, where light is confined by a solid core with a higher refractive index than the cladding, and photonic bandgap PCFs, where light guidance is achieved through photonic bandgap effects created by the periodic cladding structure [6].

1.9.3 Hole Arrangement and Geometry

The size and arrangement of air holes in PCFs strongly influence dispersion, birefringence, and light confinement. Geometries such as hexagonal, square, and

hybrid arrangements allow tailoring PCFs for enhanced sensing performance and selectivity [8].

1.9.4 Properties of PCFs

PCFs exhibit high light confinement, tunable dispersion, controllable birefringence, and low transmission loss. These properties make them ideal candidates for advanced optical sensing and fiber-based SPR systems [9].

1.9.5 Use of PCFs in Sensing

PCFs are widely adopted in sensing applications due to enhanced light–analyte interaction, high sensitivity, compatibility with plasmonic materials, and compact fiber-based sensor designs. These advantages make PCF-based sensors highly promising for biomedical, chemical, and environmental monitoring [10].

1.10PCF-Based SPR Sensors

1.10.1 PCF-SPR Sensors

PCF-based SPR sensors integrate photonic crystal fiber structures with surface plasmon resonance to achieve highly sensitive refractive index detection. Engineered air-hole geometries enhance evanescent field interaction and phase matching, resulting in superior performance compared to conventional fiber-based sensors. PCF-SPR sensors support label-free, real-time detection across biological, chemical, and environmental applications [11].

1.10.2 PCF-SPR Sensor Shapes

PCF-SPR sensors are fabricated in various configurations, including circular, D-shaped, and side-polished structures. D-shaped and side-polished designs significantly enhance evanescent field exposure and plasmonic coupling efficiency, resulting in improved refractive index sensitivity compared to circular designs[12].

1.10.3 Gold Coating

Gold is the most commonly used plasmonic material in PCF-SPR sensors due to its chemical stability, biocompatibility, and strong plasmonic response. The thickness and uniformity of the gold layer play a critical role in determining resonance characteristics and sensor sensitivity, making coating optimization essential for high-performance sensing [13].

1.10.4 Important Parameters

Key performance parameters of PCF-SPR sensors include wavelength sensitivity, amplitude sensitivity, resolution, and figure of merit (FOM). Optimization of air-hole diameter, pitch spacing, and metal layer thickness enables extremely high sensitivity and excellent sensing performance for biomedical, chemical, and environmental applications [15].

1.11 Applications of PCF-SPR Sensors

1.11.1 Biosensing

PCF-SPR sensors enable label-free, real-time detection of biomolecules such as glucose, proteins, DNA, and cancer biomarkers. Enhanced light–matter interaction results in high sensitivity and specificity, making these sensors suitable for medical diagnostics and early disease detection[17].

1.11.2 Chemical Sensing

PCF-SPR sensors are highly effective for detecting toxic chemicals and industrial pollutants by monitoring refractive index changes. Their tunable structure allows real-time, high-sensitivity chemical detection for environmental safety and industrial monitoring [19].

1.11.3 Environmental Monitoring

PCF-SPR sensors are widely used for detecting water pollutants, heavy metal ions, and environmental contaminants. Their real-time and label-free detection capability

makes them valuable tools for environmental quality assessment and public safety [20].

1.11.4 Medical Diagnostics

PCF-SPR sensors are increasingly applied in medical diagnostics for early disease detection and biomarker monitoring. Optimized fiber geometry and plasmonic coatings enable precise resonance responses and accurate quantification of biological analytes, supporting point-of-care and clinical applications [22].

- **The wide range of applications of SPR biosensors highlights their importance in several modern technological fields, particularly in healthcare and smart monitoring systems.**

1.12Contribution to Medical Field and IoT

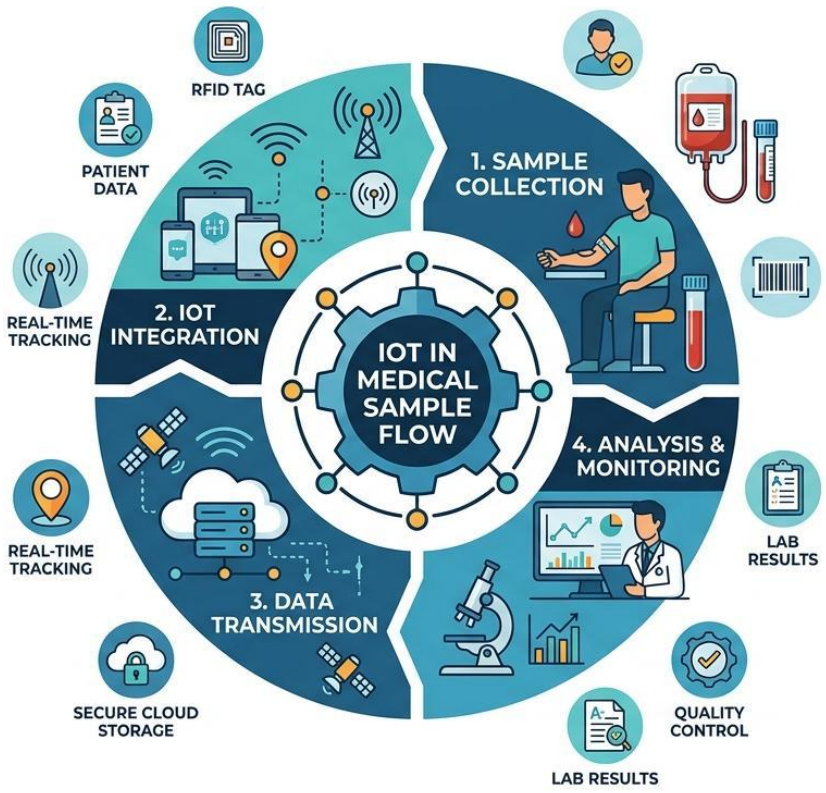

Figure1.5 : IoT-enabled biosensing system for continuous patient monitoring[22]

Surface Plasmon Resonance (SPR) biosensors play an important role in modern medical diagnostics and Internet of Things (IoT) based healthcare systems. These sensors enable rapid and highly sensitive detection of biological molecules, which helps in the early diagnosis of diseases and monitoring of patient health conditions. In addition, integrating SPR biosensors with IoT technologies allows real-time data collection and remote health monitoring. This integration improves healthcare services by providing continuous monitoring, fast response, and efficient medical decision-making[22].

Chapter 2 : Literature Review

2.1 Introduction

This chapter reviews the key previous studies related to optical sensors, particularly PCF– based SPR sensors, to understand the scientific developments that led to modern sensor designs. By examining different approaches and identifying their limitations, this chapter highlights the research gap that the proposed sensor aims to address. Overall, it provides a concise scientific background that supports the need for the design presented in this project.

2.2 Development of Fiber Based SPR Sensors

2.2.1 Transition from Conventional SPR Systems to Fiber-Based Sensors

Early surface plasmon resonance (SPR) sensors were primarily based on prism-coupled configurations, such as the Kretschmann arrangement, which required bulky optical components, precise alignment, and complex experimental setups. These systems were sensitive to mechanical vibrations and not suitable for compact or portable sensing applications [23]. To overcome these limitations, researchers introduced fiber-based SPR sensors, replacing prism-based configurations with optical fibers. Fiber-based SPR sensors offer compact size, mechanical flexibility, ease of integration, and the capability of remote sensing, making them more practical for real-world applications. The use of optical fibers also simplified the excitation of surface plasmons and enabled direct interaction between the guided light and the plasmonic layer [20]. The emergence of photonic crystal fibers (PCFs) further accelerated the development of fiber-based SPR sensors due to their unique microstructured geometry, which allows strong light confinement and enhanced evanescent field interaction with the sensing medium .

2.2.2 Challenges Addressed by Fiber-Based SPR Sensor Research

Although fiber-based SPR sensors significantly improved over conventional systems, early designs still faced several challenges. A major research focus was improving sensor sensitivity by enhancing the coupling between the core-guided

mode and the surface plasmon polariton (SPP) mode Various fiber geometries, including D-shaped fibers, dual-core fibers, and PCF-based structures, were proposed to strengthen light–matter interaction . Another critical challenge was expanding the refractive index (RI) detection range, as many early sensors were limited to narrow RI intervals, restricting their applicability to 18 diverse biological and chemical analytes.Researchers also aimed to improve sensor stability and durability by selecting suitable plasmonic materials, with gold being widely preferred due to its chemical stability and resistance to oxidation.Additionally, reducing fabrication complexity and improving tolerance to structural variations became important objectives to ensure reliable sensor performance and practical manufacturability [18].

2.3 Review of Previous PCF-SPR Designs

In 2023(PCF-SPR sensor with Au/TiO_2 layers improves cancer detection sensitivity[24].

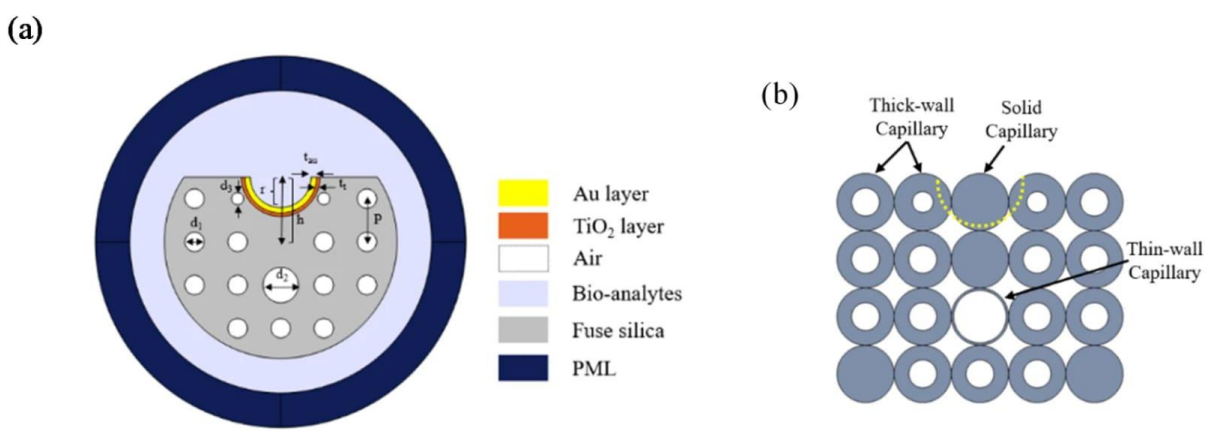

Figure2.1: Cross-sectional view of (a) proposed micro channel-based D-shaped biosensor, and (b) the stacked structure of the proposed PCF with solid, thick-walled, and thin-walled capillaries[24]

The single-side D-channel design limits multi-directional sensing capability and reduces the effective sensing area. Additionally, there is a lack of experimental validation on diverse cancer samples and insufficient long-term stability analysis[24].

In 2022 (Proposes a single-core PCF-SPR sensor with gold and V_2O_5 layers for high sensitivity and stability[25].

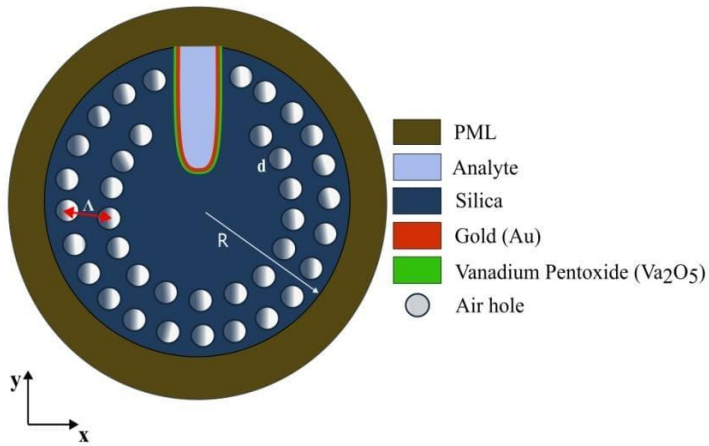

Figure2.2:Cross-sectional view of the proposed sensor design showing the material layers and structural components[25]

The single-core configuration limits simultaneous multi-sample detection. The study is mainly simulation-based and lacks practical fabrication and experimental validation. Additionally, wide-range sensing performance is not fully explored[25].

In 2021(D-shaped PCF with gold grating for enhanced SPR sensitivity using FEM[26].

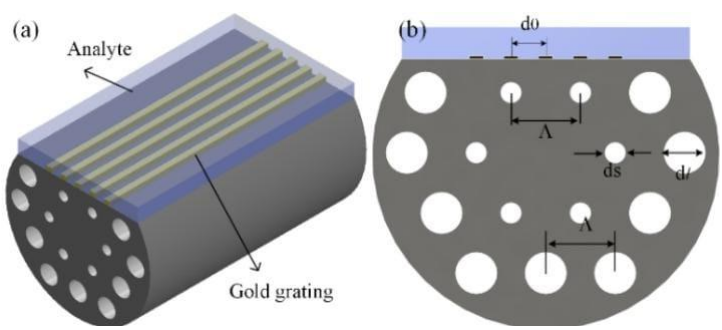

Figure2.3: schematic of proposed PCF-SPR sensor (a) 3D view (b) cross section of sensor[26].

The D-shaped structure restricts the sensing region to one side, reducing interaction between light and analyte. It also lacks dual-side enhancement and real biological sample validation[26].

In 2022(Four-channel gold PCF SPR sensor with dual polarization for higher sensitivity [27].

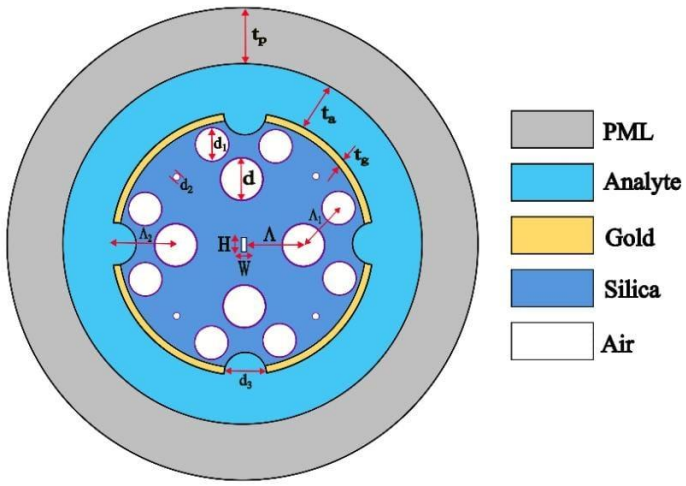

Figure2.4: schematic two-dimensional layout of the proposed SPR sensor[27]

The design involves complex and costly fabrication processes. It mainly focuses on sensitivity without addressing wide sensing range or long-term stability [27].

In 2023(PCF SPR sensor with TiO_2/gold layers optimized using deep learning [28].

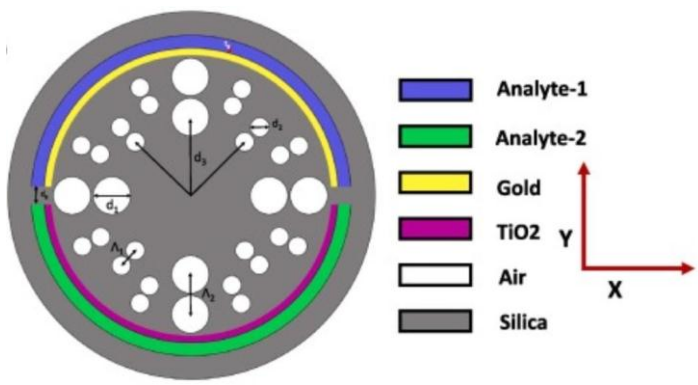

Figure2.5: Cross section of proposed PCF-SPR sensor[28].

Although deep learning improves prediction accuracy, the study lacks sufficient experimental validation with real biological samples and does not address long-term stability [28].

In 2020(Review of fiber optic SPR sensors covering principles and structural designs[29].

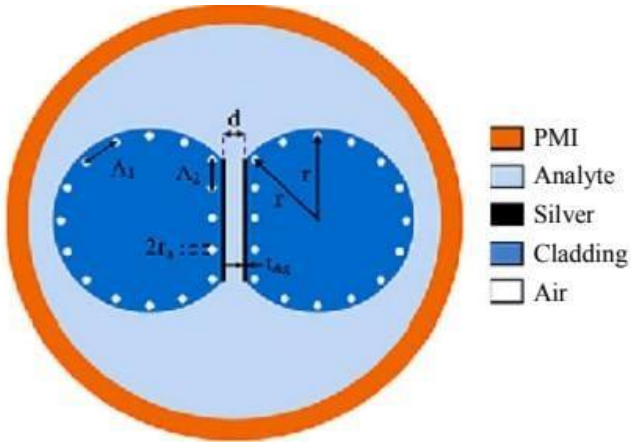

Figure2.6: symmetric double D-shaped PCFs-SPR sensor cross-section[29]

This is a review paper that does not introduce a new sensor design or experimental implementation. It also lacks practical solutions for improving sensing range and stability[29].

In 2019(PCF SPR with Ti/Ag/graphene layers for enhanced biosensing[30].

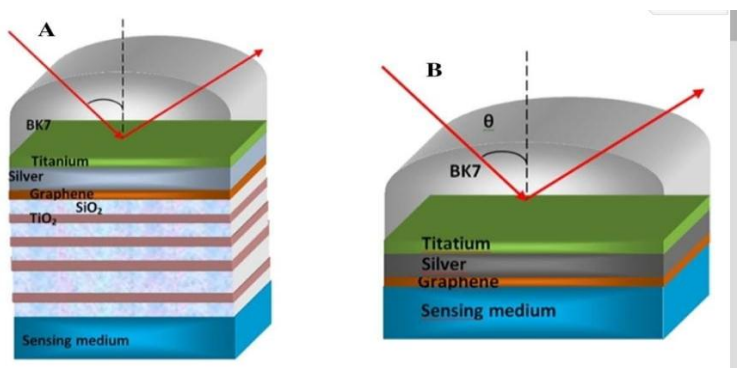

Figure2.7: Schematic diagram of suggested SPR-PhC multilayered configuration with N pairs of SiO2- TiO2layers and B the configuration of SPR graphene-based sensor with titanium-silver[30]

The work is purely theoretical and lacks experimental validation. It does not address real-world applicability or long-term performance [30].

In 2021(Review of fiber-optic SPR biosensor designs and developments [31].

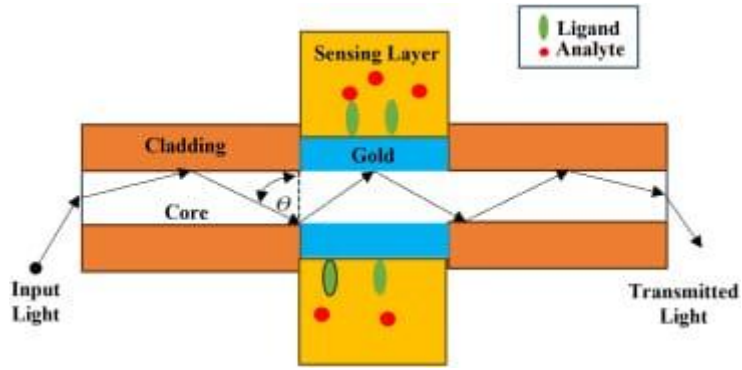

Figure2.8: schematic diagram showing the interactions between the plasmonic optical fiber biosensor and external analyte[31].

This study is a review and does not provide a novel design or experimental validation. It lacks practical strategies for real-world biosensing applications [31].

In 2022(PCF SPR optimized for sensitivity using ML and algorithms [32].

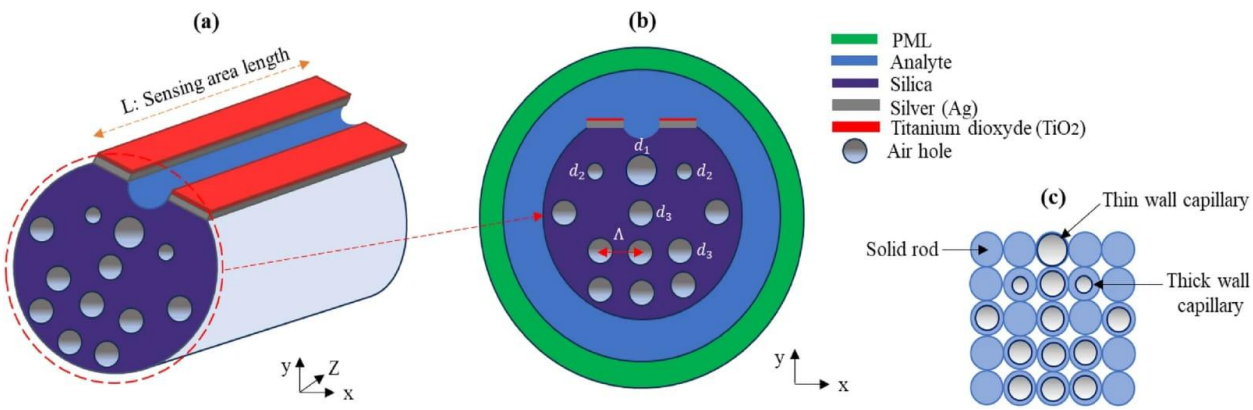

Figure2.9: Illustration depicting the proposed Dual-core PCF sensor configuration (a), the x–y cross-sectional view (b), and the stacked preform of the fiber (c)[32].

The study focuses on computational optimization without practical fabrication or experimental validation. It does not consider dual-side sensing or wide-range detection [32].

In 2018(Review of PCF SPR designs, principles, and biosensing applications [33].

This is a review paper with no novel experimental design. It lacks focus on practical implementation, long-term stability, and wide sensing range [33].

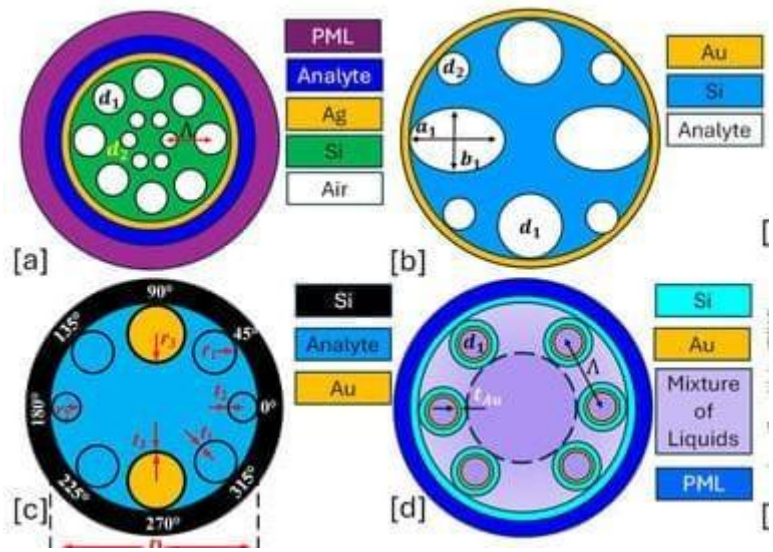

Figure2.10:Cross-section schematic of a photonic crystal fiber (PCF) SPR biosensor showing the metal coating and air-hole structure for sensing applications [33].

2.4 Current Project

In 2023 Roni etal., [34]

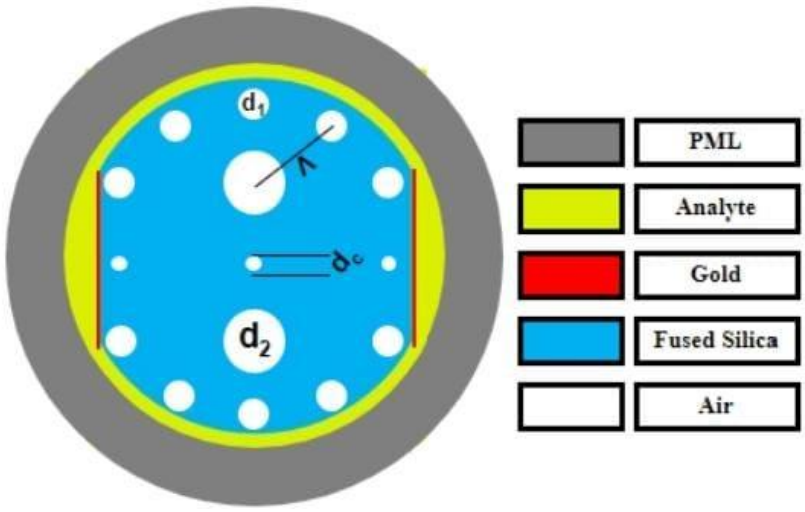

Figure2.11: Cross section of the sensor [34].

The proposed study presents a dual-side polished surface plasmon resonance (SPR) biosensor designed to achieve a wide sensing range with enhanced light–matter interaction. The sensor structure is based on an optical fiber in which two opposite sides are polished, allowing direct exposure of the evanescent field to the surrounding analyte medium. A thin gold layer is coated on the polished surfaces to support surface plasmon excitation, which is highly sensitive to refractive index (RI) variations.The performance of the sensor is analyzed using numerical simulation techniques to evaluate key parameters such as wavelength sensitivity, confinement loss, and detection range. The dual-side polished configuration significantly increases the interaction area between the guided light and the analyte, resulting in improved sensing capability compared to conventional single-side polished designs.

Results indicate that the sensor provides a relatively wide refractive index detection range while maintaining stable and reliable performance. Although its sensitivity is moderate compared to more complex PCF-based SPR sensors, the proposed design offers a simpler structure and easier fabrication process, making it highly suitable for practical implementation.

Ref	Core type	RI(RIU)	Sensitivity (nm/RIU)	stability	Fabrication complexity
[24]	single-core	1.25-1.43	47,000	medium	high
[25]	single-core	1.33-1.39	6000	medium	medium
[26]	single-core	1.36-1.38	3340	low	medium
[27]	single-core	1.36-1.42	21,250	medium	high
[28]	single-core	1.32-1.42	77,500	medium	high
[29]	various	1.36-1.38	3000	low	medium
[30]	single-core	1.33-1.39	6000	medium	medium
[31]	various	1.34-1.40	4700	low	medium
[32]	single-core	1.33-1.39	4800	Low	medium
[33]	various	1.33-1.42	4000	low	medium
[34]	Multicore/multi channel	1.21-140	5000	high	low-medium

Table 2.1 Comparative analysis of the proposed sensor with latest works

2.7 Conclusion

Although some previous studies reported higher values in sensitivity and refractive index range, it is important to note that most of these results were obtained under ideal simulation conditions and highly complex designs, which may limit their practical applicability.In contrast, the proposed Dual-Side Polished SPR Biosensor with Wide Sensing Range offers a balanced performance between sensitivity, detection range, and practical feasibility. The dual-side polishing enhances the interaction between light and the plasmonic surface, providing reliable sensitivity while maintaining a simple and robust structure.Moreover, many previous works focused primarily on maximizing sensitivity, often neglecting fabrication simplicity and operational stability. The proposed sensor, however, achieves competitive performance with a simpler, more feasible design, making it suitable for real fabrication and experimental validation.

Additionally, the multi-channel capability and improved stability further strengthen the applicability of the proposed sensor in real biosensing applications, where accuracy and reproducibility are critical .Therefore, the proposed biosensor can be considered a well-balanced solution that combines good performance with practical feasibility, making it more suitable for real-world applications compared to many previous designs that focused solely on theoretical values.

Chpater3: Design, Simulation, and Performance Analysis of the PCF-SPR Biosensor

3.1 Introduction

This chapter presents the structural design, theoretical analysis, and numerical simulation of a photonic crystal fiber (PCF)-based surface plasmon resonance (SPR) biosensor. The design is based on the research work titled “Dual-Side Polished SPR Biosensor with Wide Sensing Range.”The geometrical structure, material composition, and sensing mechanism of the biosensor are described in detail. In addition, the influence of key structural parameters on the overall performance is analyzed. Important performance metrics such as confinement loss, sensitivity, and figure of merit (FOM) are also introduced and discussed.Furthermore, numerical simulations are performed using COMSOL Multiphysics based on the finite element method (FEM), while MATLAB is utilized to process the obtained data and plot the corresponding curves. The performance of the original sensor design is evaluated in terms of confinement loss, resonance wavelength shift, sensitivity, and figure of merit.In addition, the improved design is analyzed and compared with the original structure to highlight the achieved enhancement in sensor performance.

3.2 Structure of Photonic Crystal Fiber (PCF)

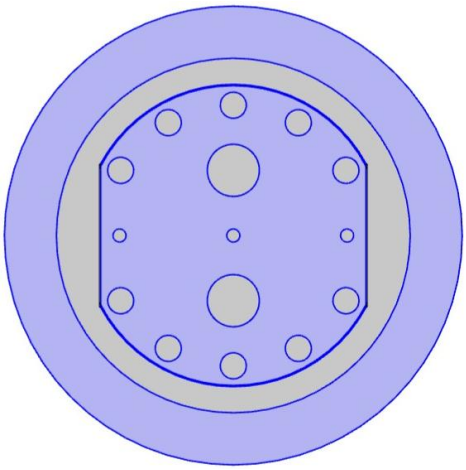

Figure3.1:Cross-sectional structure of the photonic crystal fiber showing the core and air-hole arrangement

The presented biosensor is constructed using a photonic crystal fiber (PCF), which consists of a solid silica core surrounded by multiple rings of air holes arranged in a periodic hexagonal lattice. This structure enables strong confinement of light within the core while allowing controlled interaction with the external environment.The periodic arrangement of air holes reduces the effective refractive index of the cladding region, ensuring efficient guiding of light through the core [34].

3.3 Geometry of the Original Sensor Design

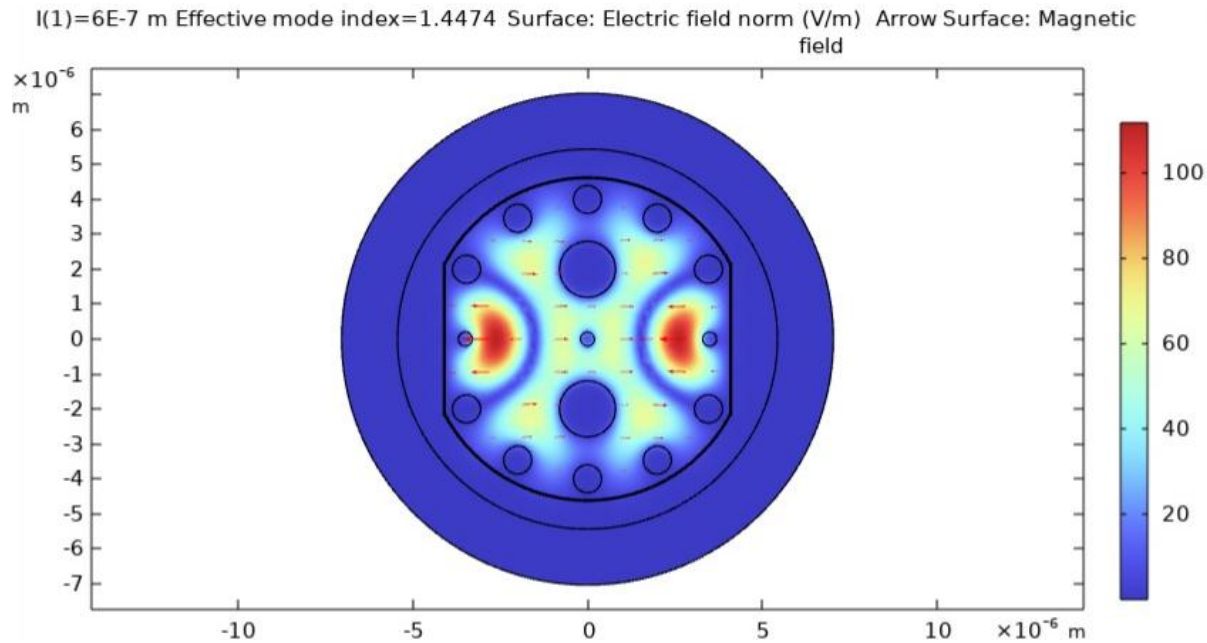

Figure3.2: Cross-sectional geometry of the dual-core PCF-SPR sensor

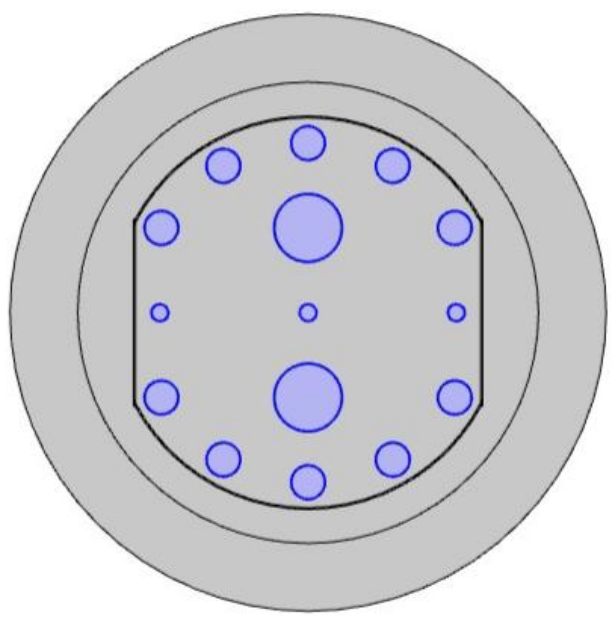

Figure3.3: Air-hole arrangement showing pitch (Λ) and air-hole diameter (d)

The sensor adopts a dual-side polished D-shaped PCF structure with a twin-core configuration. The presence of two cores enhances mode coupling, which significantly improves the interaction between the guided light and the plasmonic surface.The fiber consists of five rings of air holes arranged in a hexagonal structure.

The geometrical parameters are defined as follows:

Number of air-hole rings: 15

Pitch (Λ): 2 μm

Air-hole diameter (d): (d1= 0.8, d2= 1.6, dc= 400nm)

Ratio (d/Λ): 0.6

The pitch (Λ) represents the center-to-center distance between adjacent air holes, while the air-hole diameter (d) determines the size of each hole.

The pitch and air-hole diameter play a critical role in determining the confinement of light within the core. Increasing the air-hole diameter enhances light confinement, while changing the pitch affects the effective refractive index and the coupling efficiency between modes.

Compared to conventional single-core PCF-SPR sensors, the proposed twin-core structure provides stronger mode interaction, which results in improved sensing performance[34].

3.4 Dual-Side Polishing and Plasmonic Layer Configuration

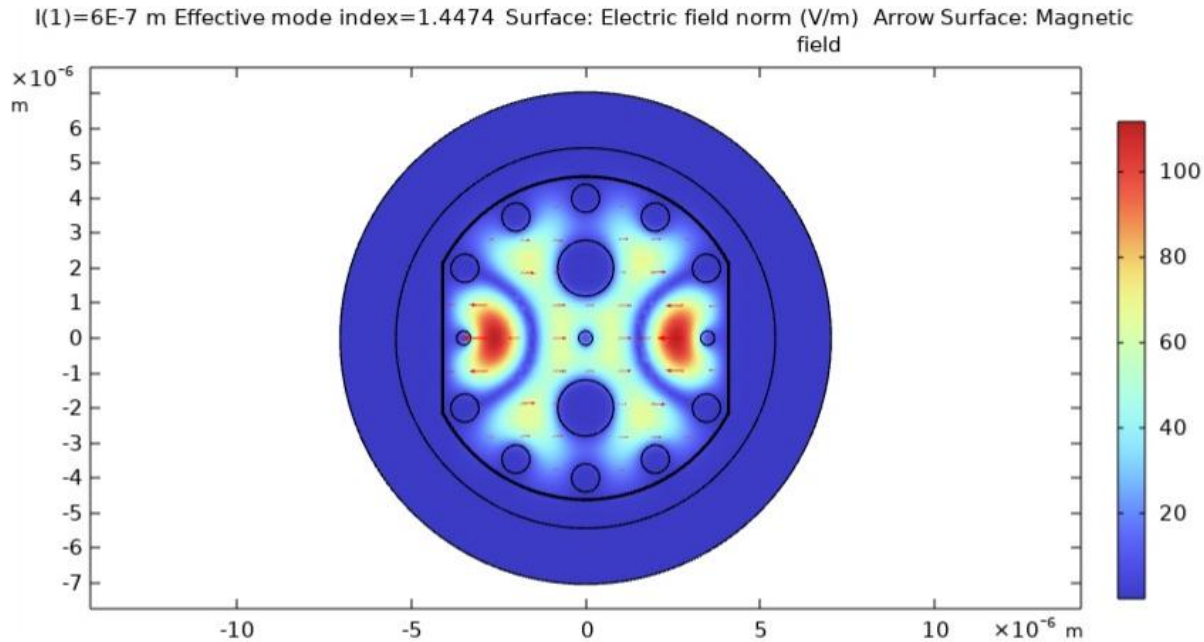

Figure3.4: Dual-side polished structure of the PCF sensor

To excite surface plasmon resonance, the fiber is polished from two opposite sides, forming a D-shaped structure. This dual-side polishing brings the evanescent field of the guided light closer to the outer surface, enabling interaction with the surrounding analyte.

A thin layer of gold (Au) is deposited on the polished surfaces, which acts as the plasmonic material responsible for SPR excitation.

The material composition of the sensor is summarized as follows:

Core: Silica (SiO_2)

Cladding: Air-hole structure

Plasmonic layer: Gold (Au)

Sensing medium: Analyte

Gold is selected as the plasmonic material due to its high chemical stability and strong plasmonic response in the visible and near-infrared regions. These properties make it highly suitable for SPR-based sensing applications.

The thickness of the gold layer is optimized at approximately 40 nm to achieve efficient resonance coupling.

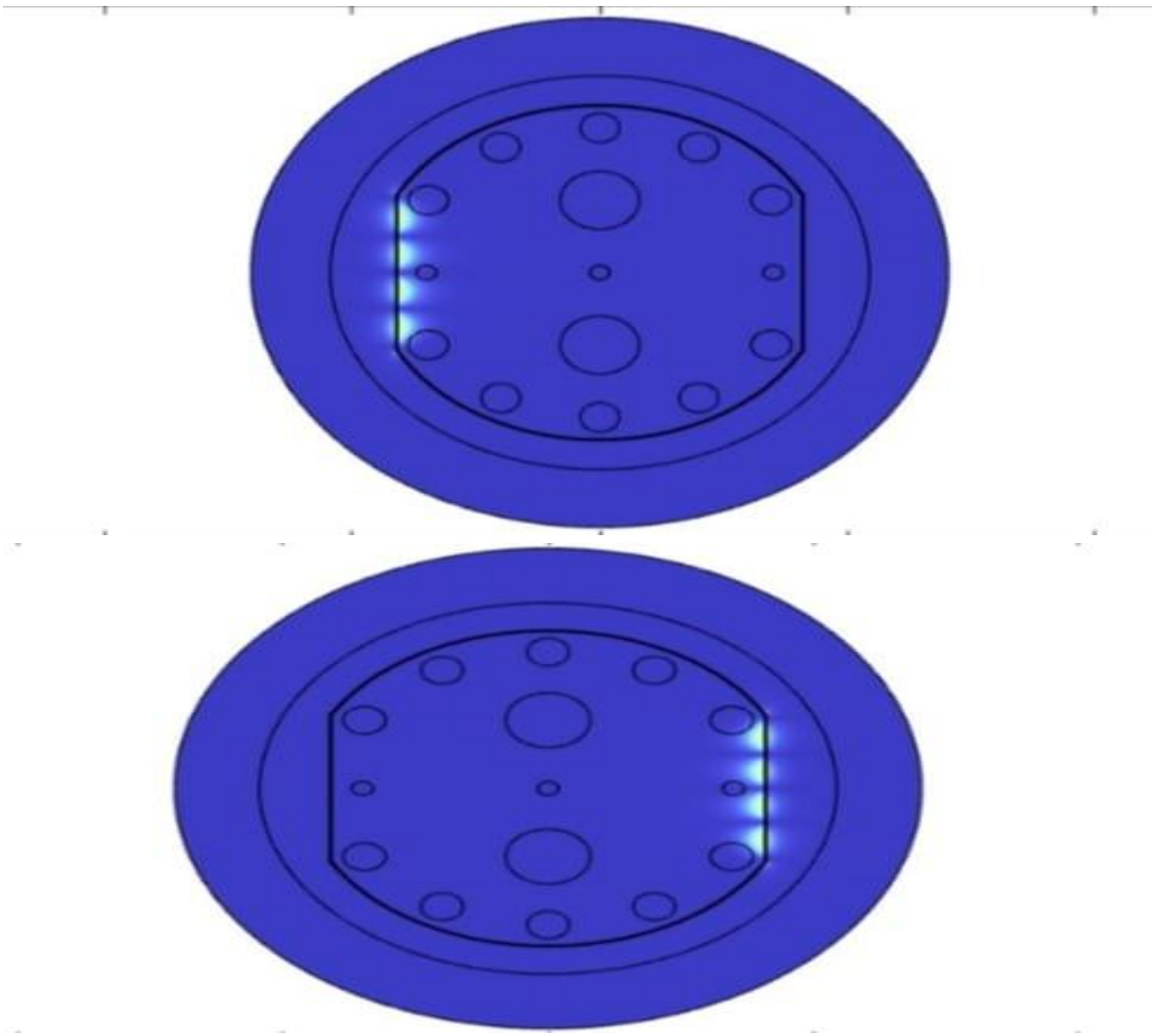

Figure3.5: Gold layer deposition on the sensing surface

3.5 Operating Principle of the SPR Sensor

The sensing mechanism is based on surface plasmon resonance (SPR), where resonance occurs when the propagation constant of the guided mode matches that of the surface plasmon mode at the metal-dielectric interface.At the resonance condition, a significant amount of optical power is transferred from the core mode to the plasmonic mode, resulting in a sharp peak in confinement loss.This resonance behavior is highly sensitive to changes in the refractive index of the surrounding analyte, which makes the sensor suitable for biosensing applications[34].

3.6 Mathematical Description of Sensor Parameters

The performance of the proposed SPR-PCF sensor is quantitatively evaluated using several key parameters derived from the modal analysis. These parameters are widely adopted in the literature and are also employed in the referenced work.

The confinement loss of the guided core mode is calculated from the imaginary part of the effective refractive index as follows:

$$\alpha_loss = 8.686 \times (2\pi / \lambda) \times Im(n_eff) \times 10^4 \text{ (dB/cm)} \quad (3.1)$$

where λ is the operating wavelength and Im(n_eff) denotes the imaginary part of the effective refractive index. A sharp peak in the confinement loss spectrum indicates strong coupling between the core-guided mode and the surface plasmon polariton (SPP) mode.

The wavelength sensitivity is defined as the shift in the resonance wavelength corresponding to a change in the refractive index of the analyte and is expressed as:

$$S_\lambda = \Delta\lambda_peak / \Delta n_a \text{ (nm/RIU)} \quad (3.2)$$

where $\Delta\lambda$_peak represents the change in the resonance wavelength and Δn_a is the change in analyte refractive index.

in addition, the amplitude sensitivity is calculated to evaluate the variation in confinement loss with respect to the analyte refractive index at a fixed wavelength. It is given by:

$$S_A = -(1 / \alpha(\lambda, n_a)) \times (\partial\alpha(\lambda, n_a) / \partial n_a)\ (RIU^{-1}) \quad (3.3)$$

The sensor resolution, which defines the minimum detectable refractive index change, can be estimated using the following relation:

$$Resolution = \Delta n_a / \Delta\lambda_min \quad (3.4)$$

where $\Delta\lambda_min$ denotes the minimum measurable wavelength shift, typically determined by the resolution of the optical spectrum analyzer[34].

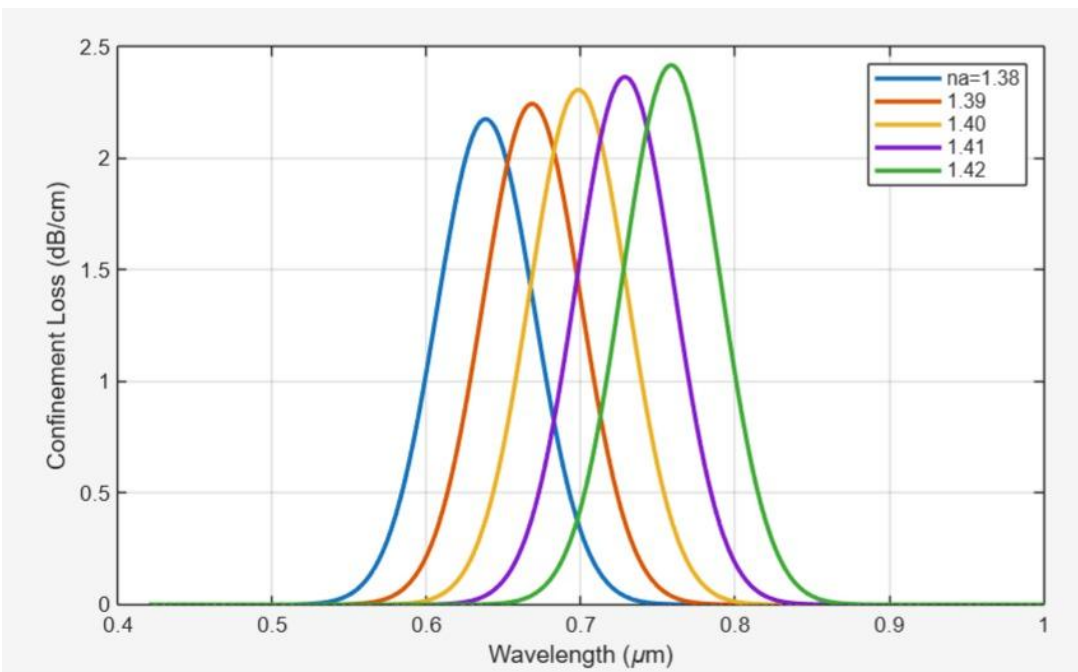

Figure3.6: Confinement loss spectrum showing resonance peak behavior

3.7 Performance Analysis of the Original Design

The performance of the original sensor is evaluated based on key parameters such as sensitivity, confinement loss, and figure of merit.

The obtained results are summarized as follows:

Wavelength Sensitivity: 5000 nm/RIU

Amplitude sensitivity: $1277\ RIU^{-1}$

RI: 2.21-1.40

FOM: 48 RIU^{-1}

Resolution: 2×10^{-5} RIU

The achieved sensitivity is mainly attributed to the dual-core structure and dual-side polishing, which enhance the interaction between the guided mode and the plasmonic layer.

These performance characteristics indicate that the proposed sensor is suitable for high-precision refractive index sensing applications.

The sensor exhibits a clear resonance peak, indicating efficient coupling between the guided mode and the surface plasmon mode [34].

3.8 Advantages and Limitations of the Original Design

The original sensor design demonstrates several advantages:

Stable resonance response

Good confinement of light

Simple structure suitable for fabrication

Compared to conventional SPR sensors, the proposed design provides improved light–matter interaction due to its structural configuration.

However, the design still has some limitations:

Moderate sensitivity compared to advanced designs

Limited sensing range

3.9 Motivation for Improvement

While the current design achieves good performance, further optimization can push the sensor toward higher efficiency.Although the proposed design demonstrates promising performance, further enhancement is still required to meet high-precision sensing applications.The current limitations in sensitivity and figure of merit motivate the need for structural optimization. Enhancing the interaction between the guided light and the sensing medium can significantly improve the overall performance of the sensor.Therefore, an improved structure is proposed in the following section to overcome these limitations.

3.10 Proposed Improved Design

To enhance the sensor performance, a structural modification is proposed by transforming the dual-core configuration into a circular sensing structure.In the original design, the dual-core arrangement provides improved mode coupling; however, the optical field remains partially confined within the cores, which limits the maximum achievable sensitivity.To overcome this limitation, the dual-core structure is replaced with a circular geometry, allowing a more symmetric and uniform distribution of the optical field around the sensing region.

The proposed modifications can be summarized as follows:

Replacing the dual-core configuration with a circular sensing region, which enhances the symmetry of the field distribution and increases the interaction between light and the analyte. Rearranging the air holes around the core to support a more uniform evanescent field distribution, leading to improved coupling with the plasmonic surface. Optimizing the interaction area between the guided mode and the gold layer by enhancing the field penetration toward the sensing surface.

These modifications significantly increase the overlap between the evanescent field and the analyte region, which is the key factor in improving SPR performance.

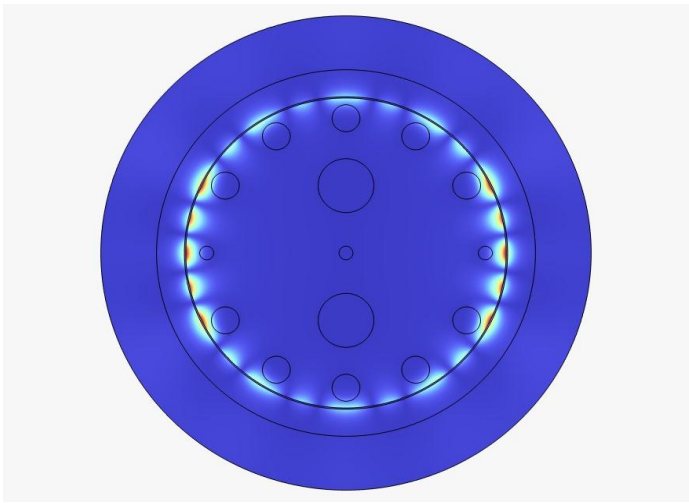

Figure3.7: Proposed modified PCF-SPR sensor design

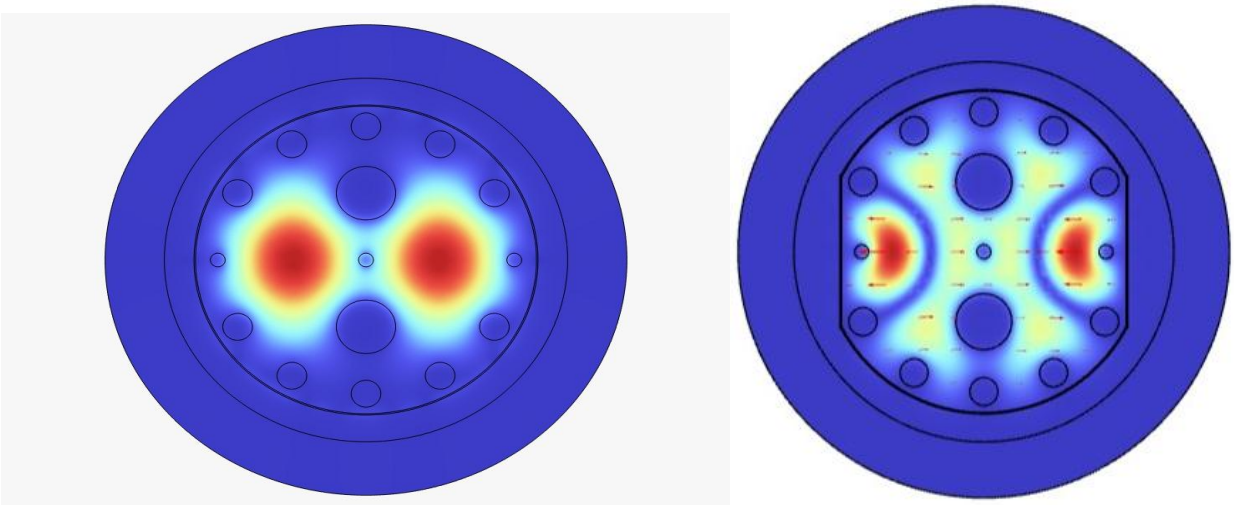

Figure3.8: Comparison between original and improved sensor structures

3.11 Expected Performance Enhancement and Trade-Off

The proposed circular structure is expected to significantly enhance the sensing performance of the biosensor.

Based on the improved field interaction, the sensor achieves:

Sensitivity increased from 5000 nm/RIU to 8000 nm/RIU

This enhancement is attributed to:

Stronger evanescent field penetration into the sensing region

Improved phase matching condition between the guided mode and plasmonic mode

Better confinement of the optical field near the sensing surface

In addition to sensitivity improvement, the modified structure is expected to produce:

Sharper resonance peak

Larger wavelength shift

Improved figure of merit (FOM)

Trade-Off Analysis

Despite the significant improvement in performance, the proposed design introduces some trade-offs:

Advantages:

High sensitivity (up to 6000 nm/RIU)

Stronger light–matter interaction

Enhanced resonance behavior

Limitations:

Possible increase in confinement loss due to stronger coupling

Increased fabrication complexity because of the circular symmetry

However, the increase in confinement loss can be considered acceptable, as it directly contributes to improving the detection accuracy and sensitivity of the sensor

3.12 Optimization Considerations

To minimize the drawbacks associated with the improved design, several optimization strategies can be considered:

Optimization of gold layer thickness

Adjustment of air-hole diameter and pitch

Selection of alternative plasmonic materials

Simulation Results and Performance Analysis

3.13 Simulation Model and Setup

The sensor geometry is implemented in COMSOL according to the structural parameters described in Chapter 3. Perfectly matched layer (PML) boundary conditions are applied to absorb outgoing waves and eliminate artificial reflections.

The main simulation parameters are:

Wavelength range: 0.8 µm to 1.8 µm

Refractive index range: 1.21 to 1.40

Gold layer thickness: 40 nm

Background material: Silica (SiO_2)

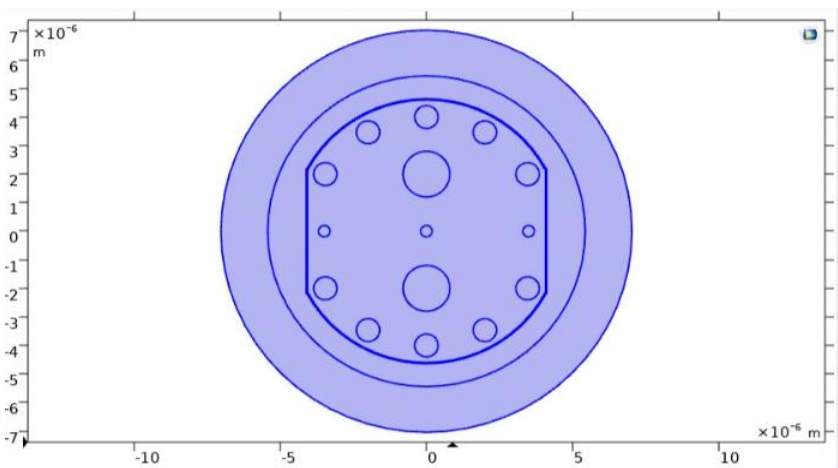

Figure3.9: COMSOL model of the dual-side polished PCF-SPR biosensor showing geometry and boundary condition

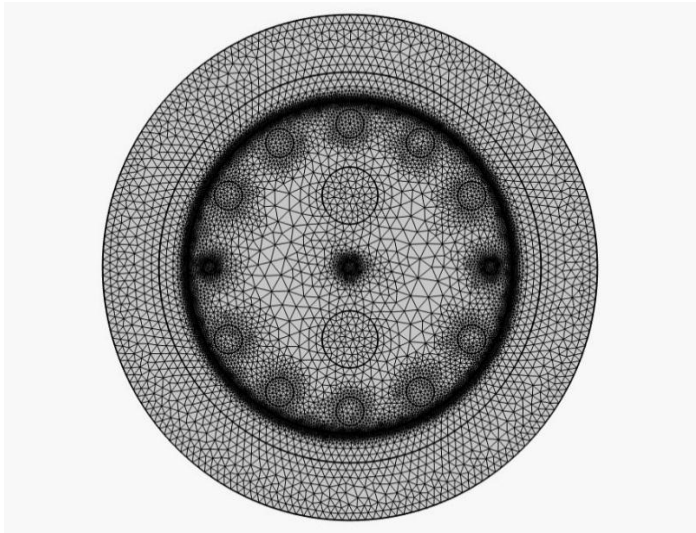

Figure3.10: Mesh distribution of the sensor structure

3.14Mode Field Analysis

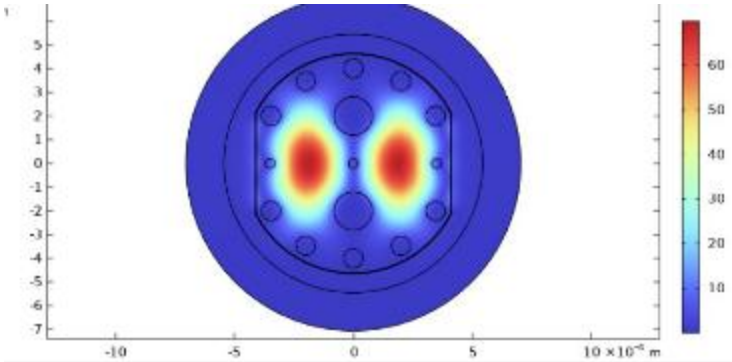

Figure3.11: Electric field distribution showing coupling between core mode and surface plasmon mods

The electric field distribution is analyzed to understand the interaction between the guided mode and the plasmonic surface.It is clearly observed that part of the guided light extends toward the metal layer, forming an evanescent field. This field is responsible for exciting the surface plasmon resonance (SPR).The stronger overlap between the optical field and the metal layer leads to stronger plasmonic excitation, which directly enhances the sensing performance[34].

3.15 Confinement Loss Analysis

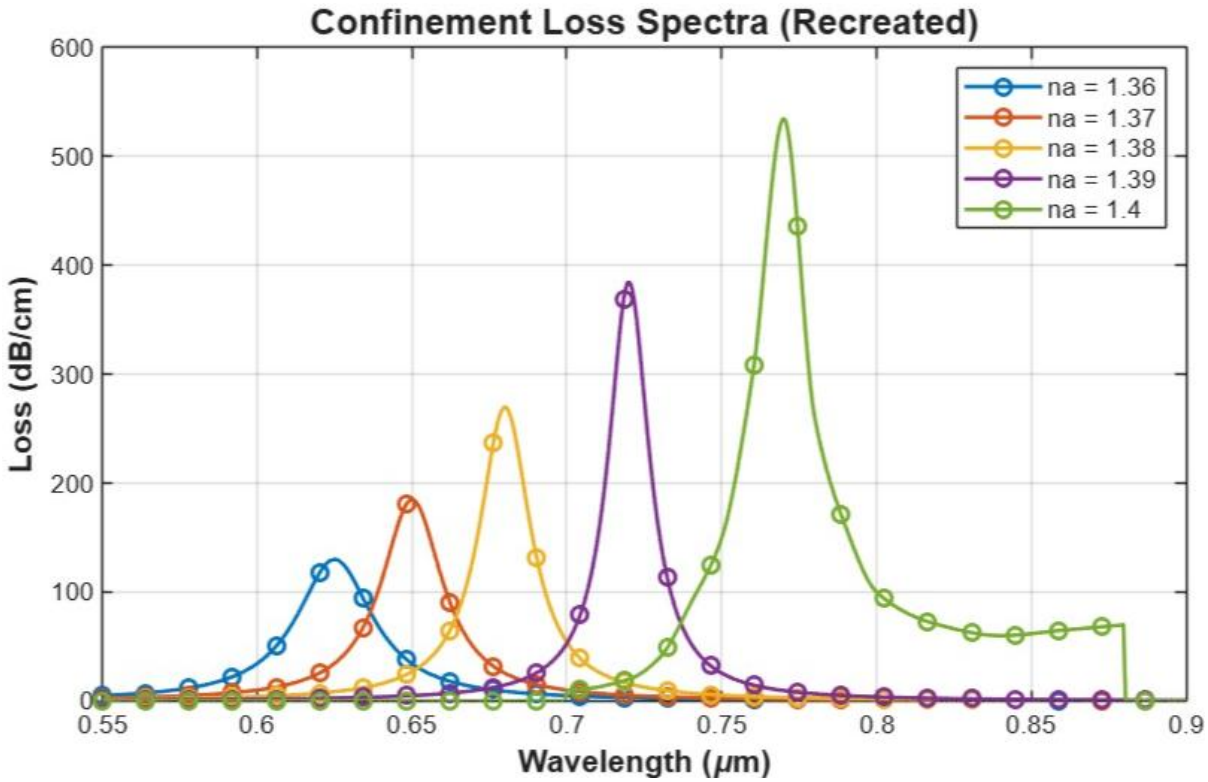

Figure3.12: Confinement loss versus wavelength for different analyte refractive indices

The confinement loss is calculated from the imaginary part of the effective refractive index and plotted using MATLAB.A sharp resonance peak is observed in the loss spectrum. This peak occurs due to the phase matching condition between the core-guided mode and the plasmonic mode.As the refractive index increases, the resonance peak shifts toward longer wavelengths. This shift occurs due to the increase in the effective refractive index, which modifies the phase matching condition between the guided mode and the surface plasmon mode.It is clear from the figure that the peak becomes more pronounced at higher refractive indices, indicating improved interaction between light and the sensing region[34].

3.16Resonance Wavelength Shift Analysis

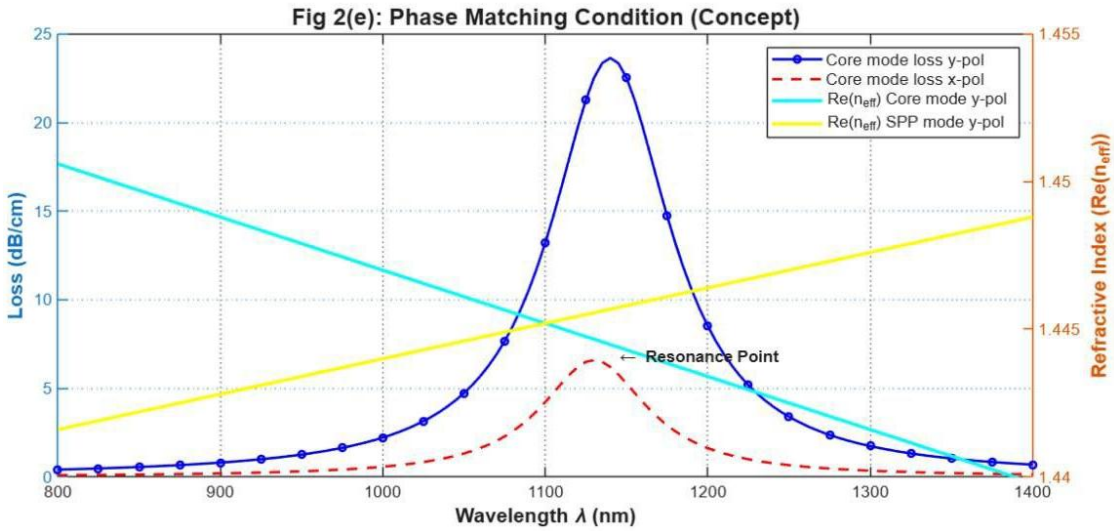

Figure3.13: Resonance wavelength shift as a function of refractive index

The resonance wavelength exhibits a linear shift with respect to the refractive index .This linear behavior is highly desirable, as it simplifies calibration and ensures accurate sensing performance .The observed shift is a direct consequence of changes in the optical path and effective refractive index inside the sensing region.

3.17Sensitivity Evaluation

The sensitivity is calculated using wavelength interrogation:

Sensitivity = 5000 nm/RIU

This high sensitivity is achieved due to the dual-side polishing and optimized interaction between the guided light and the analyte.Such sensitivity allows the sensor to detect very small refractive index variations, which is essential for biomedical applications.This sensitivity range makes the sensor suitable for detecting biological samples such as glucose levels and cancer cells, which typically fall within this refractive index range.

3.18Results of Improved Design

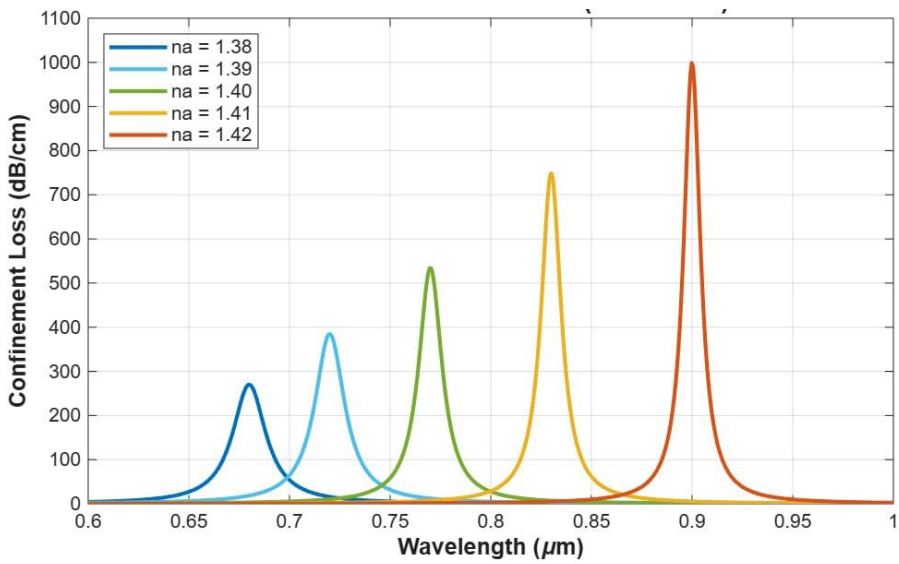

Figure3.14: Confinement loss versus wavelength for the improved design

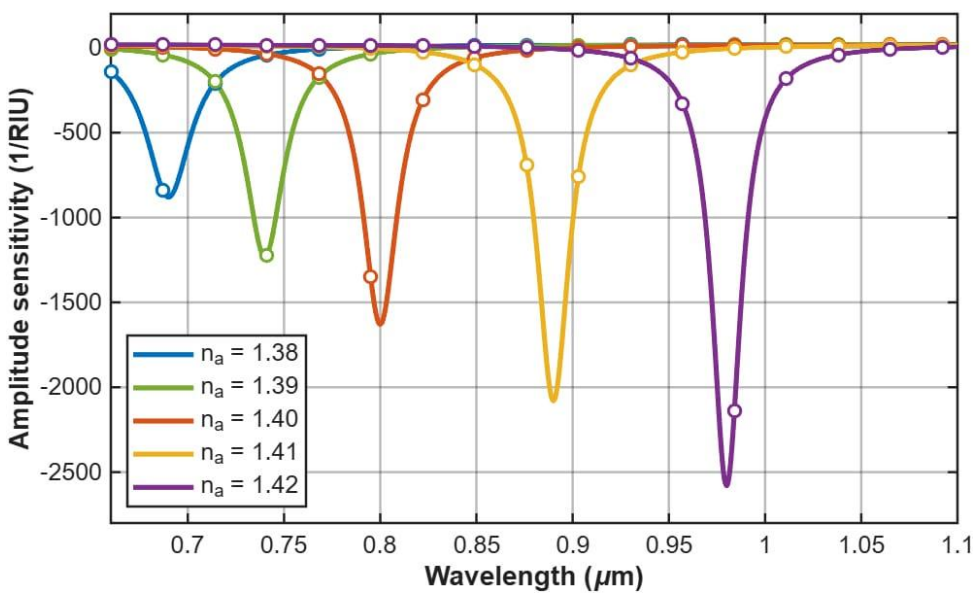

Figure3.15: Resonance wavelength shift for the improved design

Parameter	**Original Design[34]**	**Improved Design**
Sensitivity	5000nm/RIU	8000nm/RIU
Confinement Loss	550dB/cm	1000dB/cm
Fom	48 RIU^{-1}	75 RIU^{-1}
Resonance Peak	Moderate	Sharper
Wavelength Shift	Moderate	Higher
Field Distribution	Partially confined	Enhanced
Structural Complexity	Simple	Complex

Table3.1: Comparison between Original and Improved Design

The comparison clearly shows that the improved circular design achieves noticeable enhancement in sensor performance compared to the original structure.The sensitivity is increased by approximately 60%, indicating a significant improvement in the sensor's ability to detect small changes in the refractive index.This enhancement is mainly attributed to the improved electric field distribution in the circular structure, which increases the interaction between the evanescent field and the sensing region.In addition, the sharper resonance peak and the improved figure of merit (FOM) reflect higher resolution and better sensing accuracy.However, this improvement comes with a slight increase in confinement loss and higher structural complexity, which represents a common trade-off in high-performance sensor design.

- **Overall, the improved design provides a better balance between performance enhancement and structural complexity, making it highly suitable for advanced and high-precision biosensing applications.**

3.19 Conclusion

In this chapter, the structural design, theoretical analysis, and numerical simulation of the dual-side polished PCF-SPR biosensor have been comprehensively presented. The geometrical parameters, material composition, and sensing mechanism were thoroughly discussed, along with their impact on the overall sensor performance.The performance of the original design was evaluated in terms of confinement loss, sensitivity, and figure of merit (FOM), demonstrating reliable sensing capabilities. Furthermore, the improved design was analyzed and showed a significant enhancement in sensitivity and overall performance.The comparative analysis confirms that the implemented modifications effectively enhance the interaction between the guided mode and the sensing region, leading to improved sensing accuracy and efficiency. These results indicate that the optimized sensor structure is well-suited for high-precision refractive index sensing and biomedical applications.Overall, this chapter provides a solid foundation for the practical implementation of the proposed sensing system, which will be further explored in the following chapter.

Chapter 4: IoT-Based Healthcare System and Medical Application of the Proposed Sensor

4.1 Introduction

This chapter aims to transform the theoretical design of the PCF-SPR-based sensor into an integrated system applicable in the medical field through integration with IoT technologies.It explains how the signal is transmitted from the sensor, processed, and sent in real time to the user or doctor.The chapter also highlights the relationship between refractive index changes in biological samples and their use for early disease detection.The system performance is based on simulation results obtained in Chapter 4, achieving a sensitivity of approximately 8000 nm/RIU after optimization.

Finally, practical implementation challenges are discussed.

4.2 System Overview

The proposed system consists of a light source, PCF-SPR sensor, photodetector, signal processing unit, microcontroller, IoT communication module, and a cloud/user interface platform.All units work together to convert physical changes into digital signals that can be monitored remotely in near real-time.

4.3 System Block Diagram

This diagram represents the sequential flow of the signal from the light source to the final user.

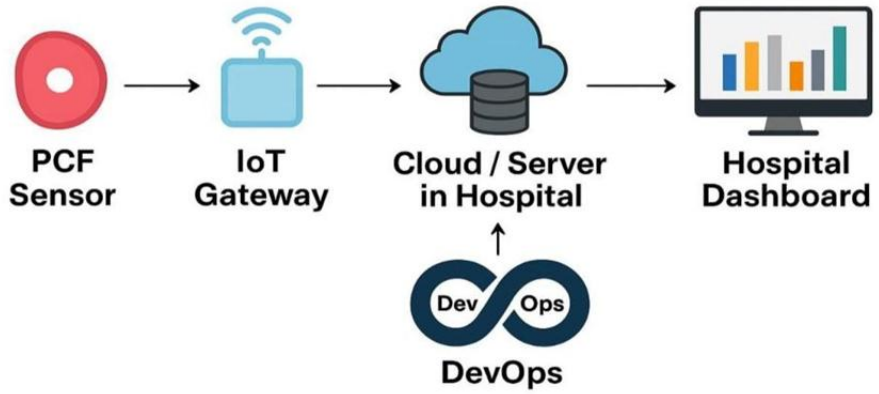

Figure 4.1: Block diagram of the proposed healthcare IoT-based system[22]

5.4 Sensor Working Principle

The sensor is based on the phenomenon of Surface Plasmon Resonance (SPR), which occurs when incident light interacts with a thin metallic layer (such as gold or silver) that is in contact with a biological sample.When light is incident at a specific angle or wavelength, the energy of photons matches the oscillation of free electrons on the metal surface. This condition leads to resonance, where strong coupling occurs between the electromagnetic field of the light and the electron oscillations.At the resonance condition, a noticeable change appears in the reflected optical signal, either as a shift in resonance wavelength or a variation in light intensity. This change is highly sensitive to small variations in the refractive index of the surrounding medium (the biological sample).In the proposed PCF-SPR sensor, light propagates through the photonic crystal fiber, and an evanescent field extends toward the sensing region where the sample is placed. Any small biological variation in the sample, such as changes in concentration or cell structure, affects the refractive index.As a result, the resonance condition is altered, leading to a measurable wavelength shift or loss change. These variations form the main sensing output, making the system highly sensitive and suitable for medical applications such as early disease detection.

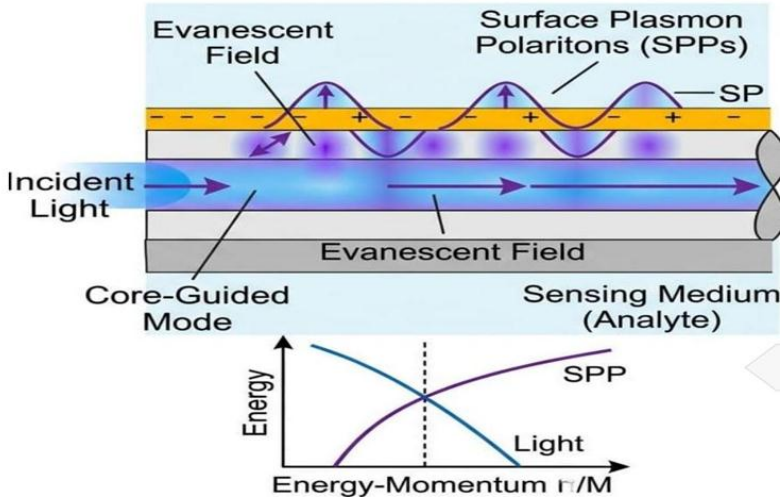

Figure4.2: Working principle of the Surface Plasmon Resonance (SPR) sensor [6]

4.5 Data Flow Analysis

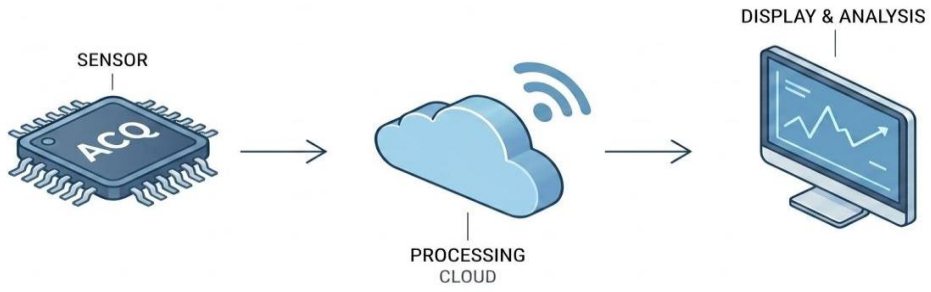

Figure4.3: Data flow process from sensor acquisition to data processing and transmission [22]

The data flow begins with the conversion of optical signals into electrical signals, followed by signal processing .Then the data is digitized, transmitted via IoT, and finally displayed to the end user or physician .The system ensures near real-time monitoring with minimal delay.

4.6 Role of Refractive Index in Disease Detection

The system does not directly detect diseases but relies on measuring changes in the refractive index of biological samples .These changes are caused by variations in biological composition due to medical conditions .Typical refractive index values of biological samples range between 1.21 and 1.40.

Table4.1: Detection of Diseases and Biomolecules using PCF-SPR Sensor

The PCF-SPR sensor detects these diseases by measuring precise changes in the Refractive Index (RI) of biological samples. The specified range (1.21 - 1.40) covers a wide clinical spectrum:Viral and Metabolic Diseases: These are concentrated in the middle range (1.33 - 1.35), including the detection of Diabetes, Anemia, and viruses such as COVID-19 and Influenza.Cancerous Conditions: These fall within the upper range (1.36 - 1.40). Cancer cells (such as Breast and Leukemia cells) possess higher optical density and a greater refractive index compared to healthy cells, allowing for highly accurate detection via Surface Plasmon Resonance.

Environmental and Gas Sensing: The lower range (1.21 - 1.30) enables the sensor to be used in environmental applications and for detecting toxic gases associated with respiratory diseases.This technology provides real-time, high-sensitivity results as soon as the sample interacts with the sensor surface and the light resonates with the metallic layer.

4.7 Conclusion

Although the PCF-SPR-based sensor demonstrates excellent performance in simulations, practical implementation remains challenging due to fabrication and operational constraints .Therefore, it is necessary to explore simpler and more practical alternative sensing approaches while maintaining high performance .The next chapter will focus on hardware implementation and alternative sensor designs suitable for real-world healthcare applications.